



Statistics of Financial Markets

Seminar 2: Classical Distributions — Normal, Student- t and Pareto



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Seminar Outline

- ▣ **Quick Review** – Key formulas, properties and comparisons
- ▣ **Multiple Choice Quiz** – 15 questions with answer options
- ▣ **True/False Questions** – 14 conceptual evaluation questions
- ▣ **Calculation Exercises** – 8 problems solved step by step
- ▣ **Python Exercises** – 6 implementation and visualization exercises
- ▣ **Discussion Questions** – 4 scenarios and critical thinking
- ▣ **AI-Assisted Exercises** – Critical thinking with LLMs
- ▣ **Summary** – Conclusions and next steps

Key Formulas to Remember

Normal Distribution:

- PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
- Skewness: $S = 0$, Kurtosis: $K = 3$

Student- t :

- Excess kurtosis: $\kappa = 6/(\nu - 4)$, $\nu > 4$
- $\nu \rightarrow \infty \Rightarrow \mathcal{N}(0, 1)$

Pareto:

- Survival: $\bar{F}(x) = (x_m/x)^\alpha$
- $\mathbb{E}[X] = \alpha x_m / (\alpha - 1)$, $\alpha > 1$

Risk Measures:

- Normal VaR: $\text{VaR}_\alpha = -(\mu + z_\alpha \sigma)$
- Normal ES: $\text{ES}_\alpha = -\mu + \sigma \cdot \phi(z_\alpha) / \alpha$
- $\text{ES} \geq \text{VaR}$ always

Model Selection:

- AIC: $-2\ell + 2k$
- BIC: $-2\ell + k \ln n$

Jarque-Bera Test:

- $JB = \frac{n}{6} \left(\hat{S}^2 + \frac{(\hat{K}-3)^2}{4} \right) \sim \chi^2(2)$

Key Concepts Summary

Concept	Key Point	When to Use
Normal	$S = 0, K = 3$; fully specified by μ, σ	Benchmark, null hypothesis
Student- t	Heavy tails, $\kappa = 6/(\nu - 4)$	Daily returns ($\nu \approx 3-8$)
Pareto	Polynomial decay $x^{-\alpha}$	Modelling extreme tails
VaR	Loss quantile	Loss limit at level α
ES	Mean of losses $>$ VaR	Coherent risk measure
AIC/BIC	Complexity penalty	Distributional model selection
Jarque-Bera	H_0 : normality ($S = 0, K = 3$)	Normality test

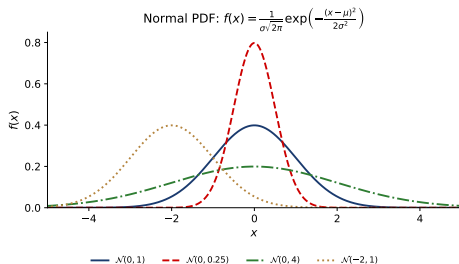
The Empirical Rule and Limits of the Normal Distribution

▣ The 68–95–99.7 Rule:

- ▣ $\Pr(\mu - \sigma < X < \mu + \sigma) = 68.3\%$
- ▣ $\Pr(\mu - 2\sigma < X < \mu + 2\sigma) = 95.4\%$
- ▣ $\Pr(\mu - 3\sigma < X < \mu + 3\sigma) = 99.7\%$

▣ Why does it fail in markets?

- ▣ The rule predicts: only 0.3% beyond 3σ
- ▣ Reality: 5σ – 10σ returns occur **tens of times** more frequently
- ▣ **Example:** “Black Monday” (1987): $-22\% \approx 25\sigma$
- ▣ Under Normal: $\Pr < 10^{-100}$ — practically impossible



Normal distribution: **symmetric, exponential tails**

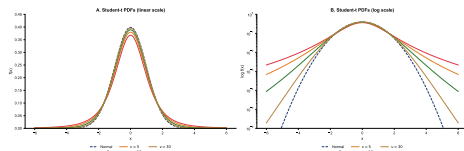
Student- t : The Effect of Degrees of Freedom ν

□ How ν controls the shape of the distribution:

ν	Variance	Kurtosis	Tails
1 (Cauchy)	∞	∞	Very heavy
3	3.0	∞	Heavy
5	1.67	9.0	Moderate
10	1.25	4.0	Light
30	1.07	3.2	$\approx$ Normal
∞	1.00	3.0	Normal

□ Financial returns: $\hat{\nu} \approx 3-8$

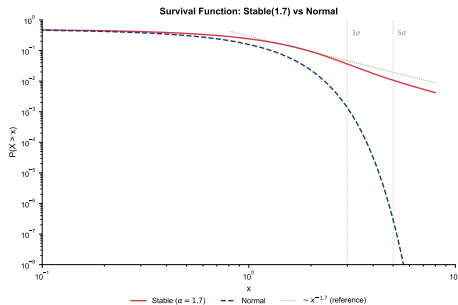
□ The smaller ν , the more extreme events



Student- t : from Cauchy ($\nu = 1$) to Normal ($\nu = \infty$)

The Pareto Distribution and the Power Law

- ▣ **Power law:** $\Pr(X > x) \sim x^{-\alpha}$
- ▣ **Polynomial decay** vs exponential (Normal)
- ▣ Tails are **much** heavier
- ▣ **Applications in finance:**
 - ▣ Extreme losses (tails of returns)
 - ▣ Wealth distribution (the 80/20 rule)
 - ▣ Trading volumes
 - ▣ Company sizes on the stock exchange
- ▣ **Key property (scalability):**
 - ▣ $\Pr(X > 2x \mid X > x) = 2^{-\alpha}$, constant in x
 - ▣ The conditional probability does not depend on the threshold!



Pareto: polynomial decay on log-log scale

Comparison: Distribution Tails

- ▣ Tail decay rate — from light to heavy:

Distribution	Tail Decay	Finite Moments	VaR 1% ($\sigma = 1\%$)	Usage
Normal	$e^{-x^2/2}$	All	2.33%	Benchmark
Student- t ($\nu = 5$)	$x^{-(\nu+1)}$	$p < \nu$	3.37%	Daily returns
Pareto ($\alpha = 3$)	$x^{-\alpha}$	$p < \alpha$	9.28%	Extreme losses
Cauchy ($\nu = 1$)	x^{-2}	No moments	–	Limiting case

- ▣ $\Rightarrow$ The heavier the tails, the more **dramatically** VaR increases
- ▣ The difference is small at 5%, but **enormous** at 0.1%
- ▣ The choice of distribution matters most in the **extreme tails**

Quiz 1: The Jarque–Bera Test

Question

The Jarque–Bera test checks:

Answer Options

- (A) Stationarity of the series
- (B) Normality of the distribution
- (C) Absence of autocorrelation
- (D) Homoscedasticity

Quiz 1: Answer

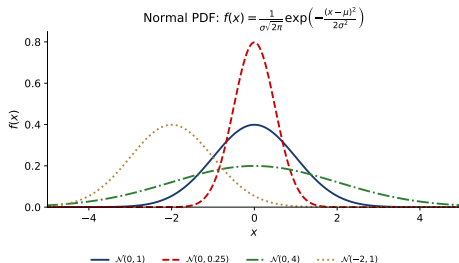
Answer: B

Normality of the distribution

□ **Explanation:**

$$JB = \frac{n}{6} \left(\hat{S}^2 + \frac{(\hat{K} - 3)^2}{4} \right) \sim \chi^2(2)$$

- H_0 : The distribution is normal ($S = 0, K = 3$)
- H_1 : The distribution is not normal
- It is based on skewness and kurtosis

Normal distribution: $S = 0, K = 3$
 SFM_ch2_normal_distribution

Quiz 2: Excess Kurtosis

Question

The excess kurtosis of the Student- t distribution with $\nu = 6$ is:

Answer Options

- (A) 1.5
- (B) 3.0
- (C) 6.0
- (D) Does not exist

Quiz 2: Answer

Answer: B

$$\kappa = 6/(6 - 4) = 3.0$$

▣ **Excess kurtosis formula:**

$$\kappa = \frac{6}{\nu - 4} \quad \text{for } \nu > 4$$

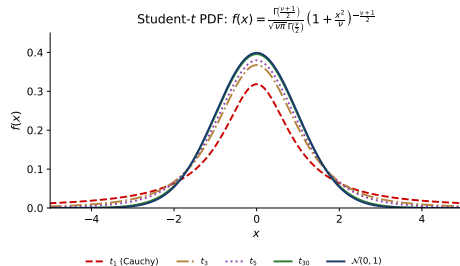
▣ Checking the options:

▣ A: $\kappa = 1.5$ would require $\nu = 8$ ✗

▣ B: $\kappa = 6/(6 - 4) = 3.0$ ✓

▣ C: $\kappa = 6.0$ would require $\nu = 5$ ✗

▣ D: Kurtosis exists for $\nu > 4$ ✗



Student-t: heavy tails, $\kappa > 0$

 SFM_ch2_student_t

Quiz 3: Normal VaR

Question

Normal VaR at 1% with $\mu = 0$, $\sigma = 1\%$ is approximately:

Answer Options

- (A) 1.65%
- (B) 2.33%
- (C) 3.09%
- (D) 3.72%

Quiz 3: Answer

Answer: B

$$\text{VaR}_{1\%} = 2.326 \times 1\% \approx 2.33\%$$

□ Normal VaR formula:

$$\text{VaR}_{\alpha} = -(\mu + z_{\alpha} \cdot \sigma)$$

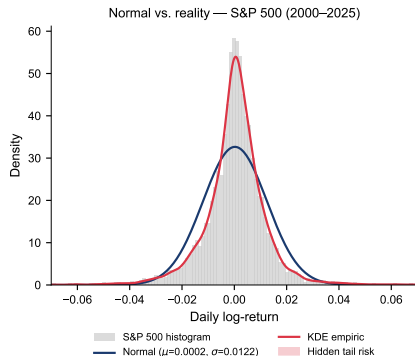
□ Key quantiles:

□ A: $z_{5\%} = -1.645$ ✗

□ B: $z_{1\%} = -2.326$ ✓

□ C: $z_{0.1\%} = -3.090$ ✗

□ D: Does not correspond to a standard level ✗



VaR: left-tail quantile of the distribution

 SFM_ch2_risk_management

Quiz 4: Student- t and Convergence

Question

When $\nu \rightarrow \infty$, the Student- t distribution:

Answer Options

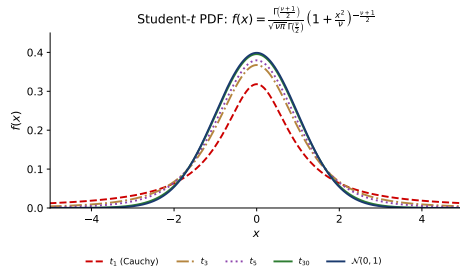
- (A) Converges to Cauchy
- (B) Converges to Normal
- (C) Variance becomes infinite
- (D) Does not converge

Quiz 4: Answer

Answer: B

Converges to the standard normal distribution

- **Distribution hierarchy:**
- $\nu = 1$: Cauchy (very heavy tails)
- $\nu = 3-8$: typical financial returns
- $\nu = 30$: nearly normal
- $\nu \rightarrow \infty$: exactly $\mathcal{N}(0, 1)$
- $\Rightarrow$ The normal distribution is a **limiting case** of the Student- t .

Student- t converges to normal as $\nu \rightarrow \infty$

SFM_ch2_student_t

Quiz 5: Expected Shortfall

Question

Which statement about ES is **true**?

Answer Options

- (A) $ES < VaR$ always
- (B) $ES \geq VaR$ always
- (C) $ES = VaR$ under normality
- (D) ES is not defined under Student- t

Quiz 5: Answer

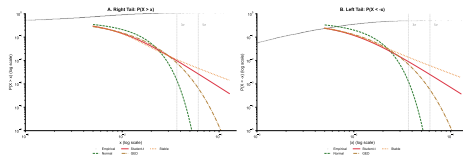
Answer: B

ES $\geq$ VaR always

Definition of ES:

$$ES_{\alpha} = \mathbb{E}[-X \mid -X > \text{VaR}_{\alpha}]$$

- ES is the mean of losses *conditional* on exceeding VaR $\Rightarrow$ it cannot be smaller than VaR.
- Under normality: $ES/\text{VaR} \approx 1.14$ at 1%.
- Under Student- t : the ratio increases as ν decreases.



ES: area under the tail beyond VaR

Quiz 6: Stylized Facts

Question

Which of the following is NOT a stylized fact of financial returns?

Answer Options

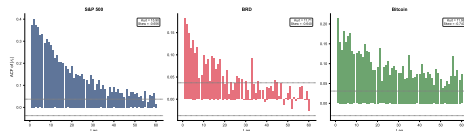
- (A) Heavy tails (leptokurtosis)
- (B) Volatility clustering
- (C) Perfect normal distribution
- (D) Negative skewness

Quiz 6: Answer

Answer: C

Real returns are NOT normally distributed

- ▣ Stylized facts (Cont, 2001):
- ▣ Heavy tails: $\hat{K} > 3$ ✓
- ▣ Volatility clustering ✓
- ▣ Negative skewness in equities ✓
- ▣ Normality: **rejected** across all markets ✗
- ▣ $\Rightarrow$ Normality is an **approximation**, not a fact.



Stylized facts: comparison of 3 assets

 SFM_ch2_case_study_2a

Quiz 7: Skewness of Returns

Question

For daily equity returns, $\hat{S} = -0.5$ suggests:

Answer Options

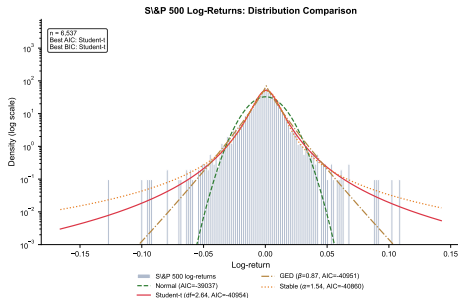
- (A) More extreme positive returns
- (B) More extreme negative returns
- (C) Perfect symmetry
- (D) Excess kurtosis

Quiz 7: Answer

Answer: B

The left tail is heavier

- ▣ Interpretation of skewness:
- ▣ $S < 0$: longer left tail (large losses) ✓
- ▣ $S > 0$: longer right tail
- ▣ $S = 0$: symmetry
- ▣ In equities: the **leverage** effect (Black, 1976) — declines are sharper than increases.



Negative skewness: heavier left tail

 SFM_ch2_distribution_comparison

Quiz 8: MLE for the Normal

Question

The MLE estimator for σ^2 in the normal case is:

Answer Options

- (A) $\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
- (B) $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$
- (C) $\frac{1}{n} \sum_{i=1}^n x_i^2$
- (D) $\frac{1}{n-2} \sum_{i=1}^n (x_i - \bar{x})^2$

Quiz 8: Answer

Answer: B

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

□ **Explanation:**

□ MLE maximizes the log-likelihood:

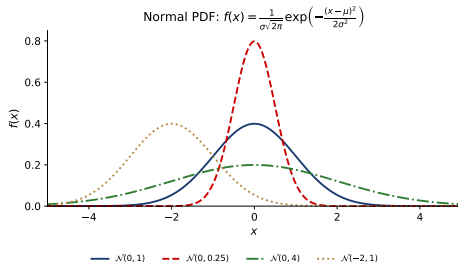
$$\ell(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

□ A: the unbiased estimator (correct, but not MLE)

✗

□ B: MLE (slightly biased) ✓

□ C: does not centre at the mean ✗



Normal: MLE coincides with method of moments

 SFM_ch2_normal_distribution

Quiz 9: The ES/VaR Ratio

Question

If VaR at 1% = 3% and ES at 1% = 4.5%, the ES/VaR ratio is:

Answer Options

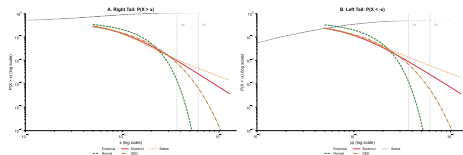
- (A) 0.67
- (B) 1.00
- (C) 1.50
- (D) 3.00

Quiz 9: Answer

Answer: C

$$ES/VaR = 4.5\% / 3\% = 1.50$$

- **The ES/VaR ratio:**
- Under normality at 1%: ≈ 1.14
- Under Student- $t(\nu = 5)$ at 1%: ≈ 1.50
- The heavier the tails, the higher the ratio
- $ES/VaR = 1.50 > 1.14 \Rightarrow$ the distribution is heavier than the normal distribution.



ES captures the severity of extreme losses

SFM_ch2_distribution_comparison

Quiz 10: BIC Penalty

Question

On a sample of $n = 5000$, BIC penalizes an additional parameter by:

Answer Options

- (A) 2
- (B) 8.52
- (C) 5000
- (D) $\ln(2) \approx 0.69$

Quiz 10: Answer

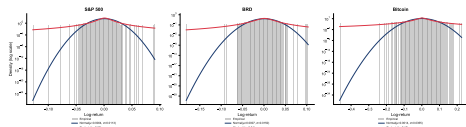
Answer: B

 $\ln(5000) \approx 8.52$ per parameter

□ AIC vs BIC comparison:

AIC : penalty = $2k$ BIC : penalty = $k \ln(n)$

- For $n = 5000$:
- AIC: +2 per parameter
- BIC: +8.52 per parameter
- $\Rightarrow$ BIC favours **more parsimonious** models.



AIC vs BIC: model selection

Quiz 11: The Pareto Distribution

Question

The survival function of the $\text{Pareto}(x_m, \alpha)$ distribution is:

Answer Options

- (A) $\bar{F}(x) = (x_m/x)^\alpha$
- (B) $\bar{F}(x) = e^{-\alpha x}$
- (C) $\bar{F}(x) = 1/(1 + x^\alpha)$
- (D) $\bar{F}(x) = (x/x_m)^{-\alpha}$

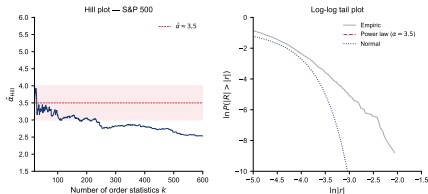
Quiz 11: Answer

Answer: A

$$\bar{F}(x) = \Pr(X > x) = (x_m/x)^\alpha \text{ for } x \geq x_m$$

□ **Explanation:**

- A: Correct — polynomial decay (power law) ✓
- B: Exponential decay (Exponential, not Pareto) ✗
- C: Log-logistic distribution ✗
- D: Equivalent to A only if $x_m = 1$ ✗
- The key: Pareto tails decay as $x^{-\alpha}$, **much slower** than e^{-x} .



Hill estimator: $\hat{\alpha}$ for Pareto tails

 SFM_ch2_extreme_value

Quiz 12: The Central Limit Theorem

Question

CLT guarantees convergence of the sample mean to Normal. Which condition is **essential**?

Answer Options

- (A) The original distribution must be normal
- (B) The variance of the original distribution must be finite
- (C) The observations must be correlated
- (D) The sample must be smaller than 30

Quiz 12: Answer

Answer: B

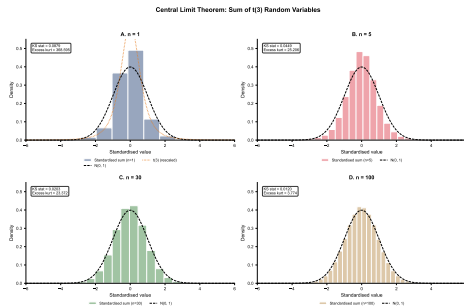
Finite variance is the essential condition

- Central Limit Theorem (Lindeberg-Lévy):

- If $X_1, \dots, X_n$ are i.i.d. with $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2 < \infty$:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

- A: The original distribution can be *anything* ✗
- B: $\sigma^2 < \infty$ is necessary ✓
- C: Observations must be *independent* ✗
- D: $n \rightarrow \infty$, not $n < 30$ ✗
- Without** finite variance $\Rightarrow$ convergence to *stable* distributions

CLT: convergence to Normal as n increases

Quiz 13: Interpreting Q-Q Plots

Question

On a normal Q-Q plot, points curving **upward** at the right end and **downward** at the left end indicate:

Answer Options

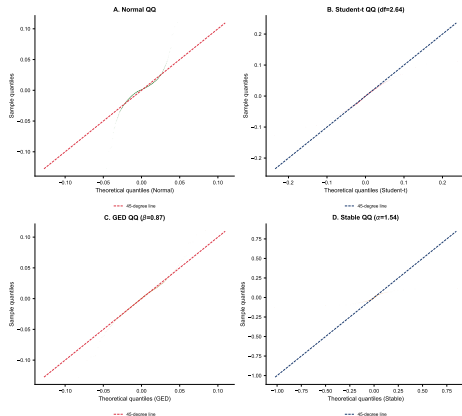
- (A) Perfect normality
- (B) Lighter tails than the normal
- (C) Heavier tails than the normal
- (D) Positive skewness

Quiz 13: Answer

Answer: C

Heavier tails than the normal distribution

- ▣ Interpreting Q-Q plots:
- ▣ Straight line: the distribution fits ✓
- ▣ S-shaped curvature: heavy tails (leptokurtosis)
- ▣ Right end ↑: larger positive extreme values
- ▣ Left end ↓: larger negative extreme values
- ▣ ⇒ Returns have more **extreme events** than the normal distribution predicts
- ▣ The Q-Q plot is *visual*, not a formal test



Q-Q plot: tail deviations indicate leptokurtosis

 SFM_ch2_distribution_comparison

Quiz 14: VaR Subadditivity

Question

VaR is not subadditive. What does this mean?

Answer Options

- (A) $\text{VaR}(A + B) \leq \text{VaR}(A) + \text{VaR}(B)$ always
- (B) $\text{VaR}(A + B) > \text{VaR}(A) + \text{VaR}(B)$ is possible
- (C) VaR cannot be computed for portfolios
- (D) VaR is always greater than ES

Quiz 14: Answer

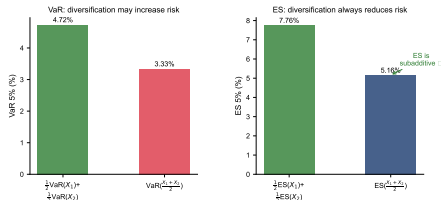
Answer: B

Diversification can *increase* VaR

- **Lack of subadditivity:**
- There exist cases where combining 2 assets **increases** total VaR
- This contradicts the diversification principle
- **Classic example:** two digital options with small loss probabilities
- ES satisfies subadditivity **always**:

$$ES(A + B) \leq ES(A) + ES(B)$$

- $\Rightarrow$ ES is a **coherent** risk measure; VaR is not

**VaR: diversification does not guarantee reduction**

Quiz 15: Moments of the Pareto Distribution

Question

If $X \sim \text{Pareto}(\alpha = 2.5)$, which moments are finite?

Answer Options

- (A) Mean only
- (B) Mean and variance
- (C) Mean, variance and skewness
- (D) All moments

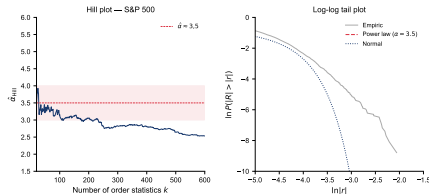
Quiz 15: Answer

Answer: B

Mean and variance, but not skewness

- ☐ **Pareto moment rule:**
- ☐ $\mathbb{E}[X^p] < \infty$ if and only if $p < \alpha$
- ☐ With $\alpha = 2.5$:
- ☐ $p = 1$ (mean): $1 < 2.5$ ✓
- ☐ $p = 2$ (variance): $2 < 2.5$ ✓
- ☐ $p = 3$ (skewness): $3 > 2.5$ ✗ infinite
- ☐ **Moment hierarchy:**

$\alpha > 1$	Finite mean
$\alpha > 2$	Finite variance
$\alpha > 3$	Finite skewness
$\alpha > 4$	Finite kurtosis

Hill estimator: $\hat{\alpha}$ determines which moments exist

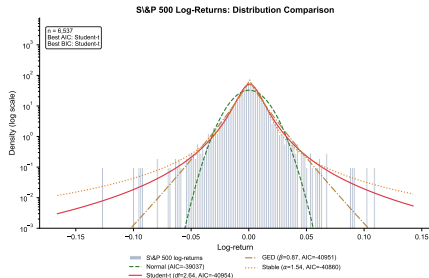
True or False? — Questions

▣ Answer with True (T) or False (F):

1. The normal distribution is fully specified by its mean and variance.
2. The Jarque–Bera test has high power on small samples ($n < 30$).
3. The excess kurtosis of the Student- t distribution exists for any $\nu > 0$.
4. Normal VaR underestimates the true risk at the 1% level in financial markets.
5. ES is a coherent risk measure, but VaR is not.
6. AIC penalizes complexity more than BIC.
7. The mean of the Pareto distribution with $\alpha = 1.5$ is finite.

True or False? — Answers

1. **T**: $\mathcal{N}(\mu, \sigma^2)$ has exactly 2 parameters
2. **F**: JB has *low* power on small samples; requires $n \geq 100$
3. **F**: Kurtosis exists only for $\nu > 4$; variance only for $\nu > 2$
4. **T**: Empirical tails are heavier than normal tails
5. **T**: VaR is not subadditive; ES satisfies the coherence axioms
6. **F**: BIC penalizes *more* ($k \ln n$ vs $2k$ for $n > 7$)
7. **T**: $\mathbb{E}[X] = \alpha x_m / (\alpha - 1)$ finite for $\alpha > 1$; with $\alpha = 1.5 \Rightarrow$ the mean exists



Normal vs Student-t vs Empirical

 SFM_ch2_distribution_comparison

True or False? — Questions (Part II)

▣ **Answer with True (T) or False (F):**

8. The Central Limit Theorem requires finite variance of the original distribution.
9. The Q-Q plot is a *formal* normality test (it produces a *p*-value).
10. Expected Shortfall is subadditive for any continuous distribution.
11. The Cauchy distribution is Student-*t* with $\nu = 1$.
12. Pareto with $\alpha = 0.8$ has a finite mean.
13. VaR at 5% is always greater than VaR at 1%.
14. Log-returns are preferred over simple returns because they are additive over time.

True or False? — Answers (Part II)

8. **T**: Standard CLT requires $\text{Var}(X) < \infty$; without finite variance $\Rightarrow$ convergence to stable distributions, not Normal
9. **F**: The Q-Q plot is a *graphical* tool, not a formal test; formal tests are JB, SW, AD
10. **T**: ES satisfies subadditivity: $\text{ES}(A + B) \leq \text{ES}(A) + \text{ES}(B)$ — a key coherence property
11. **T**: Cauchy $\equiv t_1$; it has neither a defined mean nor variance
12. **F**: The Pareto mean exists only for $\alpha > 1$; with $\alpha = 0.8 < 1 \Rightarrow$ infinite mean
13. **F**: VaR at 5% $<$ VaR at 1% (the 5% quantile is closer to the centre than the 1% quantile)
14. **T**: $r_{1 \rightarrow T} = \sum_{t=1}^T r_t$ — temporal additivity; simple returns compound multiplicatively

Exercise 1: Log>Returns and the JB Test

- ▣ **Problem:** Closing prices over 5 days: $P_0 = 100$, $P_1 = 103$, $P_2 = 99$, $P_3 = 101$, $P_4 = 105$.
- ▣ Calculate:
 1. Log-returns $r_t = \ln(P_t/P_{t-1})$
 2. Sample mean, variance, skewness, kurtosis
 3. The Jarque–Bera statistic and the conclusion at 5%

Exercise 1: Solution

1. Log-returns:

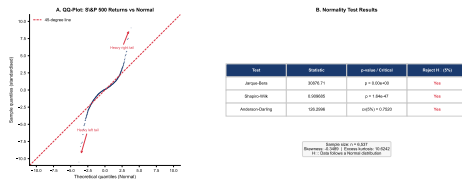
t	P_t/P_{t-1}	r_t
1	1.030	0.0296
2	0.961	-0.0396
3	1.020	0.0200
4	1.040	0.0388

2. Moments: $\bar{r} = 0.0122$, $s^2 = 0.001186$,
 $\hat{S} \approx -0.86$, $\hat{K} \approx 2.05$

3. JB: $JB = \frac{4}{6}(0.74 + 0.23) = 0.64$

$\chi^2_{0.05}(2) = 5.99 \Rightarrow$ **Do not reject H_0**

Caution: low power with $n = 4$!



Normality tests on real returns

Exercise 2: Normal VaR and ES

- ▣ **Problem:** Portfolio of 10 000 000 EUR, $\mu = 0.05\%$, $\sigma = 1.2\%$ daily.
- ▣ Calculate:
 1. Normal VaR at 5%, 1% and 0.1%
 2. ES at 1% under the normal assumption
 3. The ES/VaR ratio at 1%

Exercise 2: Solution

$$\square \text{VaR}_\alpha = -(\mu + z_\alpha \sigma) \cdot W$$

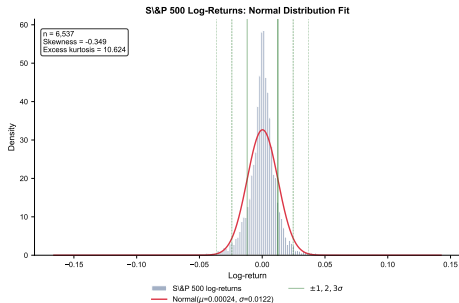
α	z_α	VaR (EUR)
5%	-1.645	192 400
1%	-2.326	274 100
0.1%	-3.090	365 800

$\square$ **ES at 1%:**

$$\text{ES}_{1\%} = -\mu + \sigma \frac{\phi(z_{0.01})}{0.01} = 3.13\%$$

$\square \Rightarrow 313\,200 \text{ EUR}$

$\square$ **Ratio:** $\text{ES}/\text{VaR} = 1.143$



VaR: left-tail quantile

Exercise 3: AIC/BIC — Normal vs Student- t

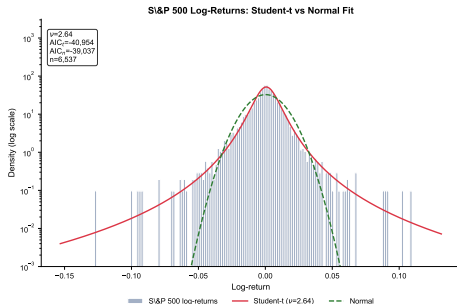
- ▣ **Problem:** On $n = 2000$ daily returns:
- ▣ Normal ($k = 2$): $\ell_N = 6421.3$
- ▣ Student- t ($k = 3$): $\ell_t = 6487.1$
- ▣ Calculate AIC, BIC and interpret.

Exercise 3: Solution

$$\square \text{ AIC} = -2\ell + 2k, \quad \text{BIC} = -2\ell + k \ln n$$

Model	k	AIC	BIC
Normal	2	-12 839	-12 827
Student- t	3	-12 968	-12 951

- $\square \Delta\text{AIC} = 129.6$
- $\square \Rightarrow$ Student- t **very significantly** preferred ($\Delta > 10$)
- $\square$ **Excess kurtosis:** $\kappa = 6/(\hat{\nu} - 4)$
- $\square$ With $\hat{\nu} = 5.2$: $\kappa = 5.0$ (much > 0)



Student- t fit: captures the tails better

 SFM_ch2_student_t

Exercise 4: Normal vs Student- t VaR

- ▣ **Problem:** Portfolio with $\mu = 0$, $\sigma = 1.5\%$ daily. Student- t with $\hat{\nu} = 5$.
- ▣ Calculate:
 1. VaR at 1% under Normal and Student- t
 2. ES at 1% under both assumptions
 3. The ES/VaR ratio for each distribution

Exercise 4: Solution

$$\square z_{1\%} = -2.326, \quad t_5^{-1}(0.01) \approx -3.365$$

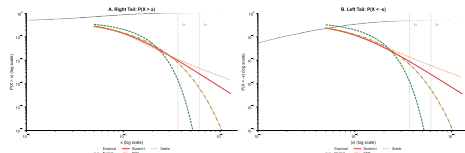
□ 1. VaR at 1%:

Model	VaR 1%
Normal	$2.326 \times 1.5\% = 3.49\%$
Student- t	$3.365 \times 1.5\% = 5.05\%$

□ 2. ES at 1%:

	ES	ES/VaR
Normal	3.99%	1.14
Student- t	7.58%	1.50

□ $\Rightarrow$ Student- t gives VaR 45% higher.



Normal vs Student- t : the difference grows in the tails

Exercise 5: Pareto Distribution — Heavy Tails

- ▣ **Problem:** The extreme losses of a portfolio follow a Pareto distribution with $x_m = 2\%$ and $\alpha = 3$.
- ▣ Calculate:
 1. The probability of a loss greater than 4%: $\Pr(X > 4\%)$
 2. The mean loss: $\mathbb{E}[X]$
 3. The variance: $\text{Var}(X)$
 4. VaR at 1% (99% quantile)

Exercise 5: Solution

1.

$$\Pr(X > 4\%) = (x_m/x)^\alpha = (2/4)^3 = 1/8 = 12.5\%$$

$$2. \text{ Mean } (\alpha = 3 > 1): \mathbb{E}[X] = \frac{\alpha x_m}{\alpha - 1} = \frac{3 \times 2\%}{2} = 3\%$$

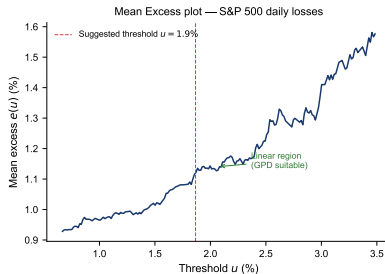
3. Variance ($\alpha = 3 > 2$):

$$\text{Var}(X) = \frac{\alpha x_m^2}{(\alpha - 1)^2 (\alpha - 2)} = \frac{12}{4} = 3\%^2$$

$$4. \text{ VaR at 1\%: } \Pr(X > q) = 0.01$$

$$q = x_m (0.01)^{-1/\alpha} = 2\% \times 100^{1/3} \approx 9.28\%$$

$$\text{Pareto VaR} = 9.28\% \text{ vs Normal } \approx 5.6\% (+66\%!)$$

Mean excess plot: linear $\Rightarrow$ Pareto tail

SFM_ch2_extreme_value

Exercise 6: Likelihood Ratio Test

- ▣ **Problem:** On $n = 1500$ daily returns of the BET index:
 - ▣ Normal ($k = 2$): $\ell_N = 5102.7$
 - ▣ Student- t ($k = 3$): $\ell_t = 5189.4$
1. Calculate the likelihood ratio (LR) test statistic
 2. The test is at significance level $\alpha = 1\%$. Do we reject H_0 : Normal?
 3. Estimate $\hat{\nu}$ from the relation $\hat{\kappa} = 4.8$ (sample excess kurtosis)

Exercise 6: Solution

1. LR statistic:

$$LR = 2(\ell_t - \ell_N) = 2(5189.4 - 5102.7) = 173.4$$

2. Test: $LR \sim \chi^2(1)$ under H_0

$$\chi^2_{0.01}(1) = 6.635$$

$$173.4 \gg 6.635 \Rightarrow \text{Reject } H_0$$

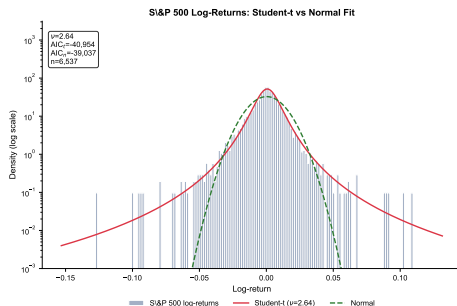
$$p\text{-value} \approx 0 \text{ — normality is clearly rejected}$$

3. Estimating $\hat{\nu}$:

$$\kappa = 6/(\nu - 4) = 4.8$$

$$\nu - 4 = 6/4.8 = 1.25$$

$$\hat{\nu} = 5.25 \text{ (typical for equities)}$$



LR test: Student-t significantly better

 SFM_ch2_student_t

Exercise 7: Cross-Asset Comparison — VaR and ES

▣ **Problem:** Three assets with estimated daily returns:

Asset	$\hat{\mu}$	$\hat{\sigma}$	$\hat{\nu}$	$\hat{S}$	$\hat{K}$
S&P 500	0.04%	1.1%	5.2	−0.3	5.5
Bitcoin	0.10%	3.8%	3.5	−0.1	8.0
Bonds (10Y)	0.01%	0.4%	7.8	0.1	3.8

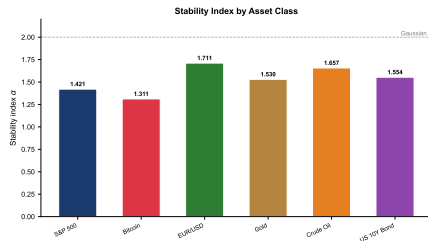
1. Calculate Normal VaR at 1% for each asset
2. Calculate Student- t VaR at 1% (using the quantile table)
3. For which asset is the Normal vs Student- t difference the largest?

Exercise 7: Solution

- Key quantiles: $z_{1\%} = 2.326$
- $t_{5.2}^{-1}(0.01) \approx 3.30$, $t_{3.5}^{-1}(0.01) \approx 4.14$,
 $t_{7.8}^{-1}(0.01) \approx 2.78$

Asset	VaR_N	VaR_t	$\Delta\%$
S&P 500	2.52%	3.59%	+42%
Bitcoin	8.74%	15.63%	+79%
Bonds 10Y	0.92%	1.10%	+20%

- $\Rightarrow$ **Bitcoin**: largest tail impact
- Reason: $\hat{\nu} = 3.5$ (very high kurtosis)
- Bonds: close to Normal ($\hat{\nu} = 7.8$)



Cross-asset: different $\hat{\alpha}$ estimates

SFM_ch2_case_study_2a

Exercise 8: Mean Excess Function and Pareto Tail

- ▣ **Problem:** Losses above threshold u have the mean excess function $e(u) = \mathbb{E}[X - u \mid X > u]$.
- ▣ For $\text{Pareto}(x_m, \alpha)$: $e(u) = u/(\alpha - 1)$.
- ▣ Data: for thresholds $u = 2\%, 3\%, 4\%, 5\%$, the observed mean excesses are:

u	2%	3%	4%	5%
$\hat{e}(u)$	1.05%	1.48%	2.10%	2.45%

1. Check whether $\hat{e}(u)$ is approximately linear in u (does it justify Pareto?)
2. Estimate $\hat{\alpha}$ from the slope of the relationship $e(u) = u/(\alpha - 1)$
3. Calculate VaR at 1% using the estimated $\hat{\alpha}$ and $x_m = 2\%$

Exercise 8: Solution

- ▣ 1. **Linearity:** ratio $\hat{e}(u)/u$:

u	$\hat{e}(u)$	$\hat{e}(u)/u$
2%	1.05%	0.525
3%	1.48%	0.493
4%	2.10%	0.525
5%	2.45%	0.490

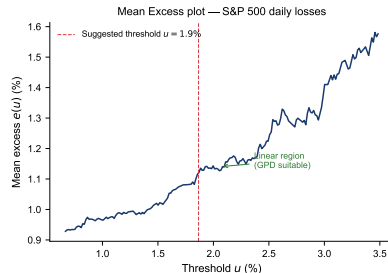
- ▣ Ratio ≈ 0.51 (linear) $\Rightarrow$ **Pareto justified**

- ▣ 2. $1/(\alpha - 1) \approx 0.51 \Rightarrow \hat{\alpha} \approx 2.96$

- ▣ 3. **VaR 1%:**

- ▣ $q = x_m \cdot (0.01)^{-1/\alpha} = 2\% \times 100^{1/2.96}$

- ▣ $= 2\% \times 4.73 \approx 9.46\%$



Linear mean excess $\Rightarrow$ Pareto tail

 SFM_ch2_extreme_value

Python Exercise 1: Download and Normality Tests

Task:

1. Download S&P 500 data for the last 5 years using `yfinance`
2. Calculate descriptive statistics (mean, std, skewness, kurtosis)
3. Apply the Jarque–Bera and Shapiro–Wilk tests

Key code:

```
import yfinance as yf
from scipy import stats
data = yf.download('^GSPC', start='2020-01-01', end='2025-01-01')
returns = np.log(data['Adj Close'] / data['Adj Close'].shift(1)).dropna()
jb_stat, jb_pval = stats.jarque_bera(returns)
```

Python Exercise 2: MLE Estimation and AIC/BIC

Task:

1. Fit Normal and Student- t via MLE
2. Calculate AIC and BIC for each model
3. Compare and select the best model

Key code:

```
mu_n, sigma_n = stats.norm.fit(returns)
df_t, loc_t, sc_t = stats.t.fit(returns)
ll_n = np.sum(stats.norm.logpdf(returns, mu_n, sigma_n))
ll_t = np.sum(stats.t.logpdf(returns, df_t, loc_t, sc_t))
aic_n = -2*ll_n + 2*2;  aic_t = -2*ll_t + 2*3
```

Python Exercise 3: VaR Comparison

Task:

1. Calculate VaR at 5%, 1%, 0.1% under Normal, Student- t and Empirical
2. Visualize the histogram with Normal vs Student- t overlay (log-scale)
3. Calculate ES via Monte Carlo under Student- t

Key functions:

- ▣ `stats.norm.ppf(alpha)` for the normal quantile
- ▣ `stats.t.ppf(alpha, df)` for the Student- t quantile
- ▣ `np.quantile(returns, alpha)` for the empirical quantile

 [SFM_ch2_distribution_comparison](#)

Python Exercise 4: Pareto — Fit and VaR

Task:

1. Extract the extreme losses (above the 95% quantile) from S&P 500
2. Fit a Pareto distribution to these losses
3. Compare Pareto VaR vs Normal VaR at 1% and 0.1%

Key code:

```
losses = -returns[returns < 0]
threshold = np.quantile(losses, 0.95)
tail = losses[losses > threshold]
alpha_hat, loc, scale = stats.pareto.fit(tail, floc=0)
var_pareto = stats.pareto.ppf(0.99, alpha_hat, loc, scale)
```

Python Exercise 5: Q-Q Plots and Visual Diagnostics

Task:

1. Construct Q-Q plots of S&P 500 returns against Normal and Student-t
2. Add a 95% confidence interval to the Q-Q plot
3. Interpret the tail deviations

Key code:

```
from scipy import stats
import matplotlib.pyplot as plt
# Normal Q-Q plot
stats.probplot(returns, dist="norm", plot=plt)
# Student-t Q-Q plot
df_hat = stats.t.fit(returns)[0]
stats.probplot(returns, dist=stats.t, sparams=(df_hat,), plot=plt)
```

Python Exercise 6: Bootstrap VaR and Confidence Intervals

Task:

1. Estimate VaR at 1% via bootstrap (10 000 resamples)
2. Calculate the 95% confidence interval for VaR
3. Compare bootstrap VaR with parametric VaR (Normal, Student- t)

Key code:

```
n_boot = 10000
var_boot = np.zeros(n_boot)
for i in range(n_boot):
    sample = np.random.choice(returns, size=len(returns), replace=True)
    var_boot[i] = -np.quantile(sample, 0.01)
ci_lower, ci_upper = np.quantile(var_boot, [0.025, 0.975])
```

Discussion 1: Normality in Financial Markets

- ▣ **Scenario:** The JB test rejects normality on S&P 500 ($p < 0.001$), but risk management still uses Normal VaR.
- ▣ **Questions:**
 1. Why is the Normal model still used if it is statistically rejected?
 2. At which VaR level (5%, 1%, 0.1%) does the difference matter most?
 3. How does the conclusion change if we use weekly instead of daily data?
 4. What are the implications for regulatory capital (Basel III)?

Discussion 2: Choosing the Degrees of Freedom ν

- ▣ **Scenario:** You estimate Student- t on 3 assets: S&P 500 ($\hat{\nu} = 5.2$), Bitcoin ($\hat{\nu} = 3.5$), Bonds ($\hat{\nu} = 7.8$).
- ▣ **Questions:**
 1. What determines the differences in $\hat{\nu}$ across asset classes?
 2. Bitcoin with $\hat{\nu} = 3.5$: which moments do *not* exist?
 3. If $\hat{\nu} < 4$, is the Sharpe ratio still valid?
 4. How does $\hat{\nu}$ change during crisis vs calm periods?
- ▣ **Hint:** Think about the relationship $\mathbb{E}[|X|^p] < \infty$ if $p < \nu$.

Discussion 3: Lessons from Financial Crises

- ▣ **Scenario:** During the 2008 crisis, the S&P 500 recorded daily returns of -9% (Oct 15) and -8% (Sep 29). Under the normal distribution with $\sigma = 1.2\%$, these events have probability $< 10^{-12}$.
- ▣ **Questions:**
 1. How many standard deviations does a -9% return represent under Normal with $\sigma = 1.2\%$?
 2. What would be the probability of the same event under Student- t with $\nu = 4$?
 3. What role did **heavy tails** play in the underestimation of risk by banks pre-2008?
 4. How would minimum regulatory capital change if banks used Student- t instead of Normal?
- ▣ **Hint:** $-9\%/1.2\% = 7.5\sigma$. Under Normal: $\Pr(Z < -7.5) \approx 3 \times 10^{-14}$

Discussion 4: Practical Trade-offs — Normal vs Student- t

- ▣ **Scenario:** You are a risk analyst. Your manager asks for a daily VaR report. You have at your disposal Normal, Student- t and the empirical method.
- ▣ **Questions:**
 1. What are the **advantages** of the Normal model? (simplicity, closed-form, CLT)
 2. When is Student- t **indispensable**? (VaR at 1%, short horizon, volatile assets)
 3. What problems does the **empirical** method have with limited data? (0.1% quantile with $n = 500$?)
 4. How do you combine all 3 approaches in a comprehensive report?

Practical Recommendation

Report VaR/ES under **all** 3 assumptions. The difference between them is informative.

AI Exercise: Critical Thinking

Prompt to test in ChatGPT / Claude / Copilot

“Download daily data for Apple stock (AAPL) from Yahoo Finance from 2015 to 2025. Calculate log-returns. Apply the Jarque-Bera and Shapiro-Wilk tests. Fit the normal and Student-t distributions via MLE. Compare AIC/BIC. Calculate VaR at 1% and 5% under both assumptions + empirical. Visualize: histogram with overlay + Q-Q plot. I want complete Python code.”

▣ Exercise:

1. Run the prompt in an LLM of your choice and critically analyze the response.
2. Does it use `stats.t.fit()` correctly (3 parameters: ν , loc, scale)?
3. Are AIC/BIC calculated correctly ($-2\ell + 2k$ vs $-2\ell + k \ln n$)?
4. Is the Q-Q plot for the t distribution or only for the normal?
5. Does it compare parametric VaR with empirical VaR? Are they consistent?

Warning: AI-generated code may run without errors and look professional. *That does not mean it is correct.*

AI Exercise 2: Comparison Across LLMs

Prompt to test in multiple LLMs

“Explain the difference between VaR and Expected Shortfall. Why is VaR not a coherent risk measure? Give a simple numerical example where VaR is not subadditive, but ES is. Show the calculation step by step.”

□ Exercise:

1. Test the prompt in **3 different LLMs** (ChatGPT, Claude, Gemini)
2. Check the numerical example: are the probabilities and values consistent?
3. Does the LLM correctly demonstrate the **lack of subadditivity** of VaR?
4. Is the formula $ES = \mathbb{E}[-X \mid -X > \text{VaR}]$ applied correctly?
5. Which LLM gives the clearest and mathematically most correct explanation?

Note: Numerical examples are the weak point of LLMs. Check every calculation step.

Key Conclusions from Today

1. **Normality is rejected** across all financial markets — the JB test confirms
2. **Student- t captures heavy tails** — $\hat{\nu} \approx 3-8$ for daily returns
3. **Excess kurtosis** $\kappa = 6/(\nu - 4)$ increases as ν decreases
4. **Normal VaR underestimates** real risk, especially at 1% and 0.1%
5. **ES $\geq$ VaR** always; the ES/VaR ratio increases with tail heaviness
6. **Pareto describes the tails** — polynomial vs exponential decay

Next Seminar

Stable Distributions

Bibliography I

Distributions and Risk Management

- ▣ Fama, E.F. (1965). "The Behavior of Stock-Market Prices," *Journal of Business*, 38(1), 34–105.
- ▣ Cont, R. (2001). "Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues," *Quantitative Finance*, 1, 223–236.
- ▣ McNeil, A.J., Frey, R. & Embrechts, P. (2015). *Quantitative Risk Management*. 2nd ed., Princeton.
- ▣ Mandelbrot, B. (1963). "The Variation of Certain Speculative Prices," *Journal of Business*, 36(4), 394–419.

Software Tools

- ▣ `scipy.stats` – Distributions, MLE, statistical tests
- ▣ `yfinance` – Financial data download
- ▣ `matplotlib`, `seaborn` – Visualization
- ▣ `numpy`, `pandas` – Computation and data manipulation

Replication Exercise: Fama (1965)

Objective

Replicate the analysis of Fama (1965) on the non-normality of stock returns, using current data.

1. Download daily returns for S&P 500 and 5 individual stocks (last 10 years)
2. Calculate descriptive statistics: mean, standard deviation, skewness, kurtosis
3. Apply the Jarque–Bera test to each series
4. Fit Normal and Student- t via MLE, compare AIC/BIC
5. Plot normal vs Student- t Q-Q plots for each series
6. **Question:** Are the conclusions of Fama (1965) still valid today?

Ref: Fama, E.F. (1965). "The Behavior of Stock-Market Prices," *Journal of Business*, 38(1), 34–105. <https://www.jstor.org/stable/2350752>

Homework: Seminar 2

Objective

Analyze the distribution of returns for 3 assets from different classes.

1. Download daily returns for: a stock index (e.g. BET), a stock (e.g. FP.RO) and a cryptocurrency (e.g. BTC-USD) — last 5 years
2. Calculate descriptive statistics: mean, standard deviation, skewness, kurtosis
3. Apply the Jarque–Bera, Shapiro–Wilk, Anderson–Darling tests
4. Fit Normal and Student- t via MLE, compare AIC/BIC
5. Calculate VaR and ES at 1% under both assumptions + empirical
6. Plot normal vs Student- t Q-Q plots for each asset
7. **Question:** Which asset deviates the most from normality? Why?

Thank you!

Questions?

Seminar materials are available at: <https://danpele.github.io/SFM/>



Quantlet



Quantinar